



Sample and Population

If you can't study the entire group of objects that you are interested in, then study just some of them.

Distinguishing Between a Sample and a Population

A **population** is an entire set of members that you want to know something about. The set can be large or small, and it can be made up of people or objects. Here are examples of populations:

- every resident in the United States
- every library card holder in San Antonio, Texas
- every lion in a nature preserve
- every apple in an orchard

A **sample** is a subset of a population. It is common to study samples when a population is very large and studying each member would be impossible, costly, or time-consuming.

For example, a science professor is considering sending some of the university's science students on a field trip to another country. He wants to know how many of the 918 science students at his university would participate in such a trip. He surveys 83 of the science students and finds that 34 of them would participate.

Who makes up the population?

The 918 science students at that university make up the population. Ideally, the professor would be able to survey each of them, but his schedule does not allow him time to survey each student, especially those who have other professors.

Who makes up the sample?

The 83 students whom the professor surveys make up the sample.

► **Think About It** Every 10 years, the U.S. Census Bureau gathers basic information on every resident in the United States. Annually, the bureau gathers more detailed information from a sample of the population.

Finding and Using Sample Means

A group of biology students is studying Mexican free-tail bats in a local cave. They capture a dozen bats from the cave, record their wingspans in centimeters, and then release them. The wingspans are 31 cm, 36 cm, 34 cm, 32 cm, 33 cm, 33 cm, 29 cm, 35 cm, 30 cm, 34 cm, 35 cm, and 30 cm. Estimate the average wingspan of all the bats in the cave.

The sample mean is a good estimate of the population mean. To find the sample mean, divide the sum of the wingspans in the sample by the number of bats in the sample.

$$\frac{392}{12} \approx 32.7$$

Based on this sample, it is fair to estimate that the average wingspan of all the bats in the cave is about 33 cm.

► **Think About It** The accuracy of an estimate based on a sample depends largely on how the sample was produced and the size of the sample.

Finding and Using Sample Proportions: Fractions

A university prepares to send 520 biology test kits to overseas students. An assistant randomly inspects 40 of the kits and finds that 5 have outdated instructions. About how many kits in the shipment have outdated instructions?

Find the proportion of test kits in the sample with outdated instructions.

$$\frac{5}{40} = \frac{1}{8}$$

Because $\frac{1}{8}$ of the test kits in the sample have outdated instructions, it is fair to estimate that $\frac{1}{8}$ of the test kits in the entire shipment have outdated instructions—that is, $\frac{1}{8} \cdot 520 = 65$ test kits.

► **Remember** You can write a proportion as a fraction, decimal, or percent.

Finding and Using Sample Proportions: Percents

Sometimes it makes more sense to write the sample proportion as a percent.

After returning from a biology field trip, a professor asks a sample of the students whether they would return the following year. The table shows the results.

Response	Frequency
Yes	21
No	11
Unsure	6

► **Q&A**

Q How do you determine the size of the sample?

A Find the sum of the frequencies.

$$21 + 11 + 6$$

A total of 80 students went on the trip. Estimate how many would say *No* if they were asked whether they would return the following year.

Find the percent of students in the sample that would not return. Divide the number who responded *No* by the total number in the sample.

$$\frac{11}{38} \approx 29\%$$

Find 29% of the population.

$$0.29 \cdot 80 = 23.2$$

Based on the sample, it is fair to estimate that about 23 of the 80 students would say *No* to returning the following year.